Beregningsteknik for Computer Ingeniører

MM 7: Complex functions

Topics:

The complex exponential function

Euler's formula

The complex logarithmic function

Complex exponential

Note, the imaginary value of *z* is also the argument of the complex exponential

- The complex exponential function definition

$$(z = x + iy);$$
$$e^z = e^{x+iy} = e^x e^{iy} = e^x(\cos(y) + isin(y))$$

…. an analytic function whose derivative is $(e^z)' = e^z$

- If we see the exponential as combining two real functions

$$f(z) = e^z = u(x,y) + iv(x,y) = e^x\cos(y) + ie^x\sin(y)$$

- … the derivative can be derived as the partial derivative wrt. *x* or y !

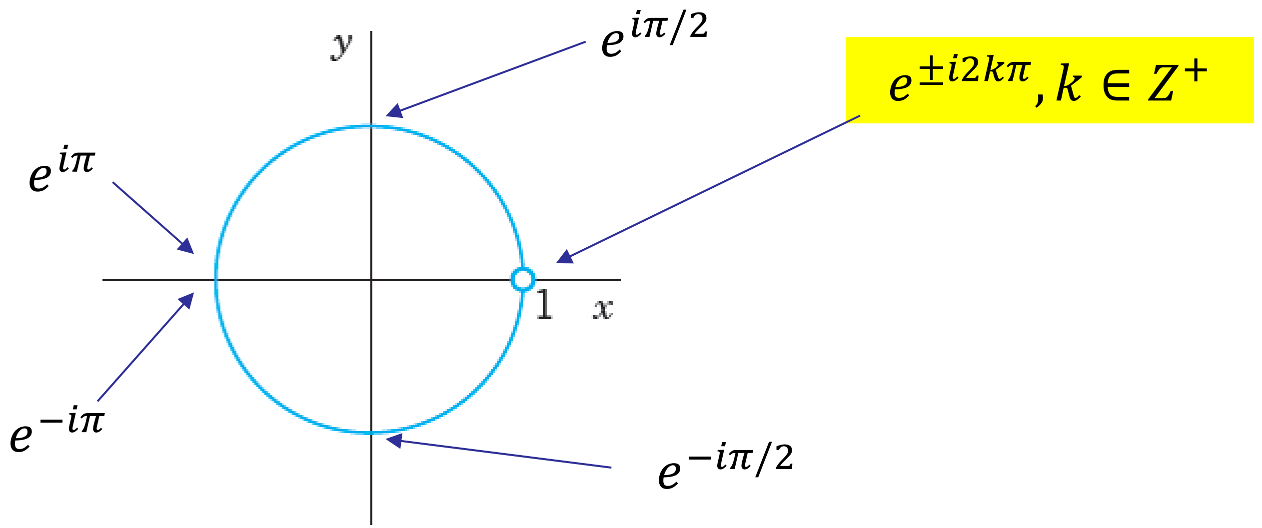
$$\frac{\partial e^z}{\partial x} = \frac{\partial u(x,y)}{\partial x} + i\frac{\partial v(x,y)}{\partial x} = \frac{\partial e^x\cos(y)}{\partial x} + i\frac{\partial e^x\sin(y)}{\partial x} \stackrel{?}{=} \frac{\partial e^z}{\partial y}$$

- … depending on how we approach *z*, but we need partial derivatives $u_x = v_y$ and $u_y = -v_x$ to be satisfied (Cauchy-Riemann equations) for the complex exponential function to be analytic.

Euler's formula

- Euler's formula

$$z = iy;\ e^z = e^{x+iy}\Big|_{x=0} = \boxed{e^{iy} = \cos(y) + isin(y)} \qquad |e^{iy}| = 1$$



- Polar form of *z (which we previously stipulated by a change of variables)*

$$r = e^x;\ z = r(\cos(\theta) + isin(\theta)) = re^{i\theta}$$

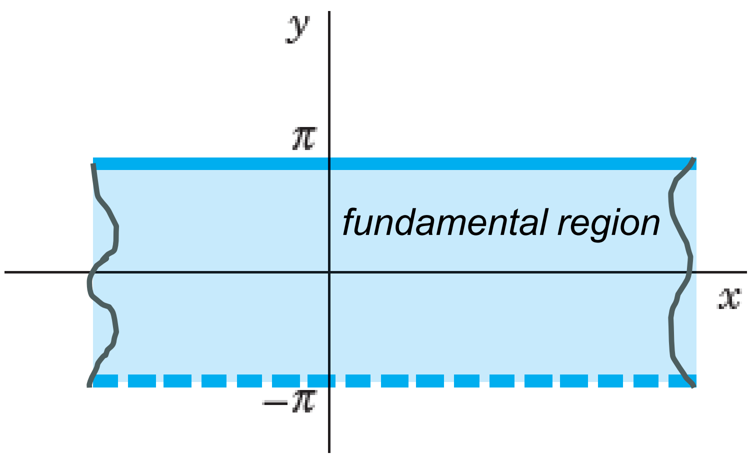
Domain of exponential

- The complex exponential is periodic

$$w = e^z = e^{x+iy} = e^{z+i2\pi}$$

The imaginary value of *z* is also the argument of the complex exponential - which is periodic with 2π

- … with all fundamental values in a horizontal strip of 2π, giving all the values that e^z can assume



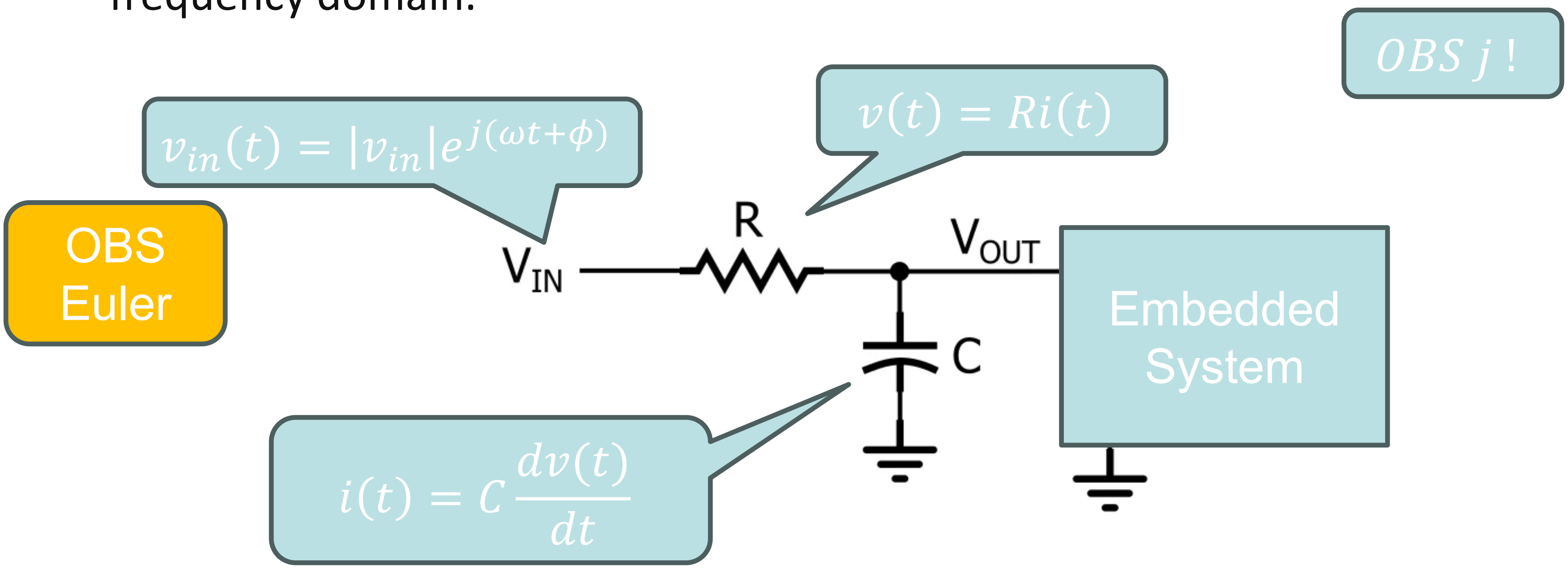
$$-\pi < y \leq \pi$$

See illustration later in connection with logarithm

Complex numbers in Computer Engineering

Signal Processing (5th semester course):

- Complex numbers are essential in Laplace/Fourier transforms, which are used to analyze and manipulate signals. This is crucial for applications like audio and image processing, where signals are often represented in the frequency domain.



Euler – extension to complex

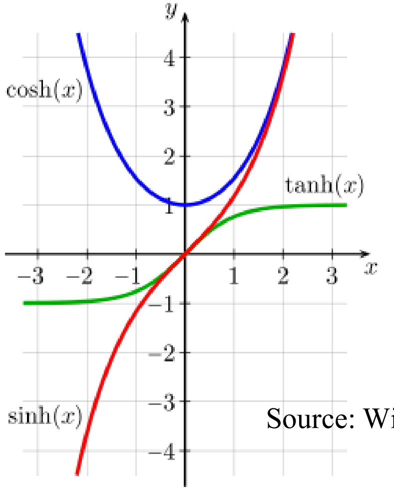
- Trigonometric functions (definitions)

$$\cos(x+iy) = \cos(z) = (e^{iz} + e^{-iz})/2$$

! $$|\cos(z)|^2 = \cos^2 x + \sinh^2 y$$

$$\sin(x+iy) = \sin(z) = (e^{iz} - e^{-iz})/(2i)$$

! $$|sin(z)|^2 = sin^2 x + \sinh^2 y$$



Source: Wiki

- Based on this, we can even define Euler's formula for complex *z*

$$\cos(z) + isin(z) = e^{iz}$$

- Similarly, the hyperbolic functions can be defined in complex *z*

$$\cosh(z) = (e^{z} + e^{-z})/2$$
$$\sinh(z) = (e^{z} - e^{-z})/2$$

Logarithmic function

- The inverse of the exponential function is the natural logarithm

Note: to be consistent with Kreyszig, now $z = e^{w}$

$$\mathrm{w} = \ln(\mathrm{z}), \mathrm{z} \neq 0; \qquad (z = x + iy = re^{i\theta})$$

$$z = e^{w} = e^{u+iv} = e^{u}e^{iv} \stackrel{\text{def}}{=} re^{i\theta} \qquad \mathrm{r} = |\mathrm{z}| = \mathrm{e}^{\mathrm{u}}; v = \theta$$

….. so we see that *arg(z)* = *θ* equals *w*'s **imaginary** part

$$\theta = \arg(\mathrm{z}) = v$$

- Hence, the complex logarithm is defined only up to a multiple of 2π !

$$w = \ln(z) = u + iv = \ln(r) + i\theta = \ln(|z|) + i\, arg(z), r > 0$$

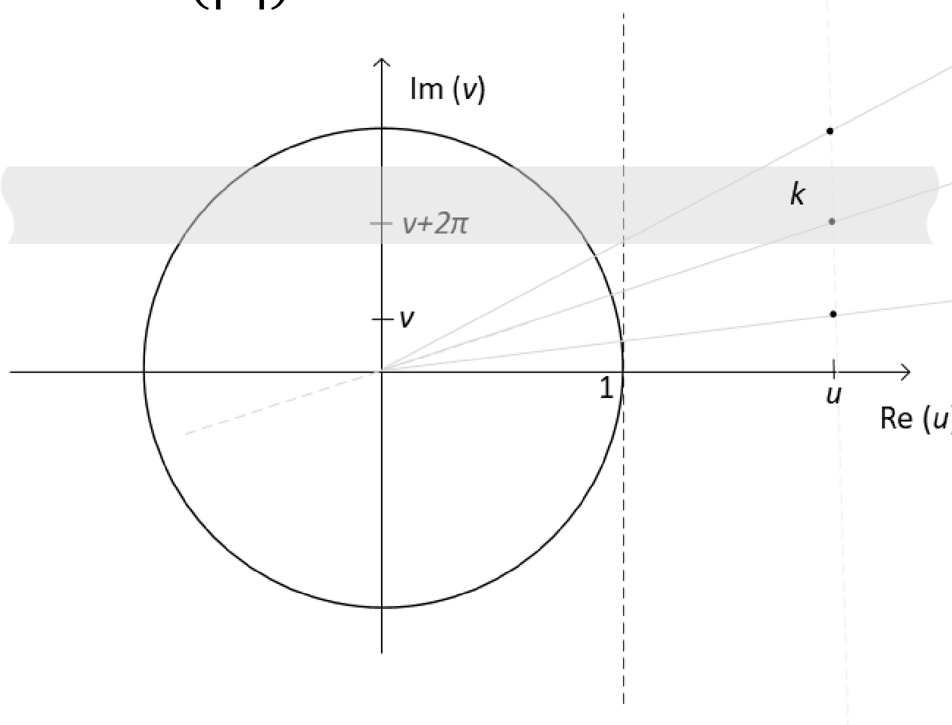
- The value corresponding to the principal value of Arg(z) is called the ***principal value (of)*** $\mathbf{Ln}(\mathbf{z})$

$$\ln(\mathrm{z}) = \mathbf{Ln}(\mathbf{z}) \pm i2n\pi, n \in Z^{+}$$

Log multi-valued

$$w = \ln(z) = u + iv = \ln(r) + i\theta = \ln(|z|) + i(arg(z) + 2\pi k)$$
$$\Downarrow$$
$$arg(w) = \mathrm{tg}^{-1}\frac{arg(z) + 2\pi k}{\ln(|z|)}$$

***Note**, the argument is always up to a multiple of 2π – which, however, is not of interest here - but the fact that EVEN WITHOUT there are infinitely many values of w for the same z, all with the same real part.*



”Complex” relations

- Some interesting observations (based on previous page)

$$e^{\ln(\mathrm{z})} = e^{\ln(|\mathrm{z}|)+\mathrm{i}\arg(\mathrm{z})} = e^{\ln(|\mathrm{z}|)}e^{\mathrm{i}\arg(\mathrm{z})} = |z|e^{i\theta} = \mathrm{z}$$

… as we know it from the real valued functions (maps to one value), but …

$$ln(\mathrm{e}^{\mathrm{z}}) = ln(|\mathrm{e}^{\mathrm{z}}|) + \mathrm{i}arg(\mathrm{e}^{\mathrm{z}}) = x + i(y \pm 2k\pi), = z \pm i2k\pi, k \in Z^{+}$$

... since

$$e^{\mathrm{z}} = e^{x+iy} = e^{x}e^{iy} \stackrel{\mathrm{def}}{=} |e^{\mathrm{z}}|e^{i\,arg(e^{\mathrm{z}})}$$

$$\ln(|e^{\mathrm{z}}|) = \ln(e^{x}) = x;\ arg(e^{z}) = y \pm 2k\pi, k \in Z^{+}$$

- Therefore, $ln(\mathrm{e}^{\mathrm{z}})$ is multi-valued (as was $ln(z)$)

General powers

- We already encountered powers of *z* in connection with the generalized version of De Moivre's formula

$$z^n = r^n(\cos(n\theta) + isin(n\theta)), n \in Z$$

… these are single-valued (cartesian form) since *n* is an integer

- If we let the exponent be $1/n, n \in N$ we get the roots (as before)

$$z^{1/n} = \sqrt[n]{z} = e^{(1/n)\ln(z)}$$

… with *n* distinct values for arguments $2\pi k/n, k \in \{0,1, \ldots n-1\}$ (determined up to multiples of 2π)

- Generalizing this to complex exponents *c* we write

$$z^c = e^{cln(z)}, z \in C \neq 0, c \in C$$

… which is (inf.) multi-valued since *ln*(*z*) is; using *Ln*(*z*) gives the principal power value.

Complex numbers in Computer Engineering

Cryptography:

- Complex numbers play a role in encryption algorithms, helping to secure communication by scrambling messages in a way that makes them unreadable to unauthorized users.
  - One method involves using complex logarithms. By transforming complex numbers into position vectors (Argand diagram), we can utilize the properties of complex logarithms to encode and decode messages.
  - This approach provides an additional layer of security by making it more difficult for unauthorized parties to decipher the message without the correct key and understanding of the complex transformations involved

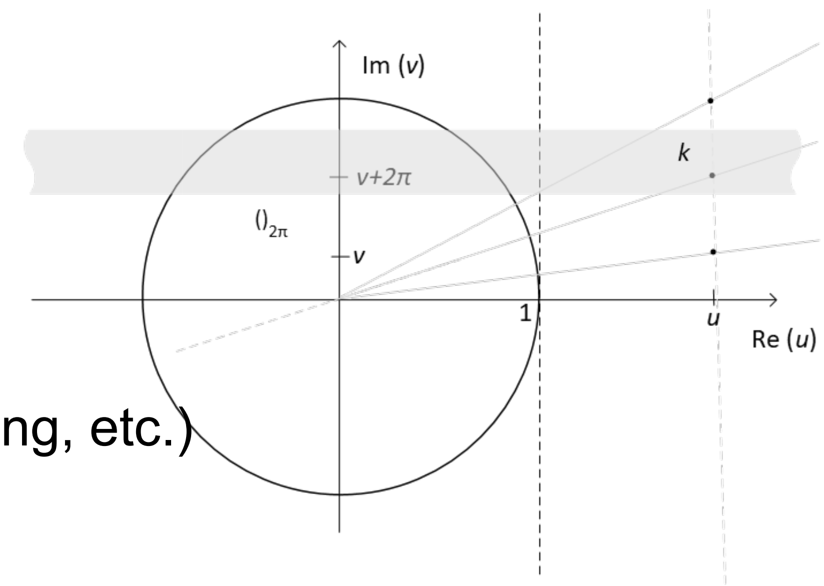
Simple encryption

Encryption

- Generate *z* from plaintext
  - Constrain *z* (imag. part) to the open *connected subset* $G \in \mathbb{C}$ (random branch *k*) – by scaling and offsetting
- Generate (encrypt) cypher $\mathrm{c} = e^{Re(z)+i\,Im(z)}$

Decryption

- Uses $\ln c$ (one to many)
- Knowing k
  - $\ln c + i2\pi k$
- Needs to be combined with other encryption techniques (mixing, shifting, etc.)
- Problems?



Complex exponents

- A more “complex” example, just for illustration and application of complex exponents, but somewhat same principles:
  - Consider $e^{aln(z)}$ where $a \in \mathbb{C}$ (as well as *z*)

$$z^a = e^{aln(z)} = e^{aln(|z|)+ia\,arg(z)} = e^{aln(|z|)+ia(Arg(z)+2\pi k)}$$

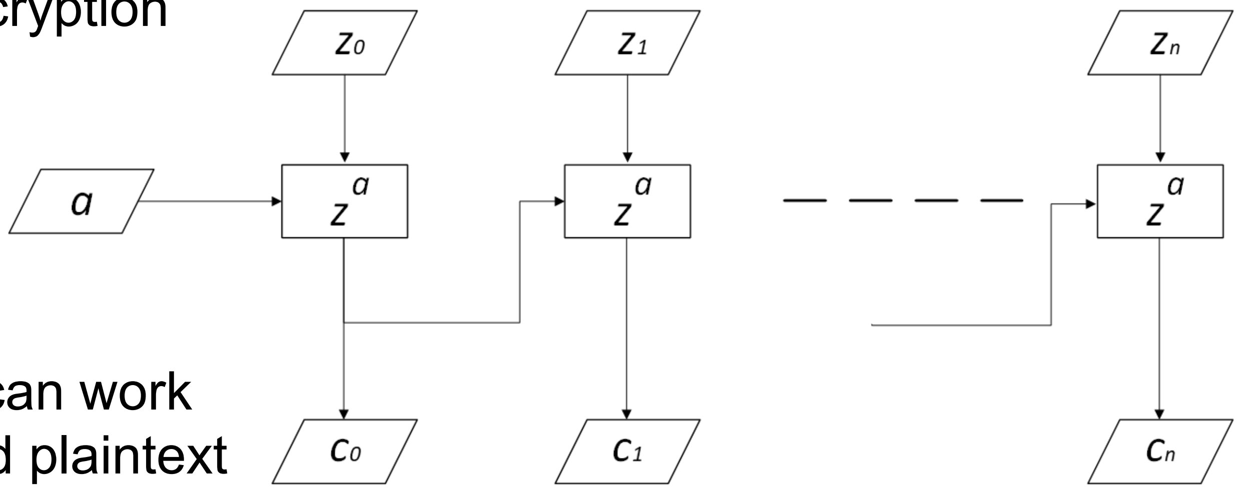
  - Generally, this covers the cases:
    - $a \in \mathbb{Z}$ (integer) → only one output value (e.g. *a* = *n*)
    - $a \in \mathbb{Q}$ (rational) → finite number of output values (e.g. *a* =1/*n*)
    - $a \in \mathbb{C}$ $(\mathbb{Q}')$ (complex/irrational, imaginary part not 0) → infinite number of output values (with different *k*)

- Having infinite output makes it harder to attack the encryption, having to search an infinite space (for possible output values with *k*).

Chain encryption

- Simplified principle, where previous complex number (a, C_o) is used as exponent a in following encryption

$$z^a = e^{a\ln(z)} = e^{a\ln(|z|)+ia(Arg(z)+2\pi k)}$$



- Knowing k, one can work backwards to find plaintext and exponents

G. Stergiopoulos, M. Kandias and D. Gritzalis, "Approaching encryption through complex number logarithms," 2013 International Conference on Security and Cryptography (SECRYPT), Reykjavik, Iceland, 2013,